

Ritwik Gupta

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CAREER OBJECTIVE

Enthusiastic Computer Science graduate with a strong foundation in Generative AI. Proven ability to develop and implement AI models, demonstrated through academic projects. Proficient in Python and key GenAI frameworks, with a keen interest in applying AI to solve real-world challenges.

EDUCATION

Pranveer Singh Institute of Technology (PSIT), Kanpur, UP

(Up to Semester 7)

- B.Tech in Computer Science (AI)

2022-2026 | Percentage: **74.09**

Methodist High School, Kanpur, UP

- Intermediate (10+2) (ISC)
- High School (ICSE)

2019-2020 | Percentage: **79**

2017-2018 | Percentage: **91.3**

TECHNICAL SKILLS

- Programming Languages:** C++, Python
- Core Competencies:** OOPs, DBMS
- Databases:** SQLite
- Vector Database:** FAISS, Qdrant, BM25 Retrieval, Reciprocal Rank Fusion (RRF), Dense Retrieval
- Deployment:** Streamlit Cloud, Python 3.10, Environment Management
- GenAI & NLP:** Hybrid Retrieval, Prompt Engineering, Semantic Search, Retrieval-Augmented Generation (RAG), Sentence Embedding

PROJECTS

AI Research Paper Copilot — Production-Grade Multi-Paper RAG System

2026

- Built a production-grade GenAI research assistant using Hybrid RAG (BM25 + Dense Retrieval + RRF), Qdrant, and Groq Llama 3.1 for citation-grounded research Q&A, literature review generation, and paper comparison across multiple academic papers.
- Engineered advanced retrieval pipeline with semantic search, Reciprocal Rank Fusion (RRF), section-aware chunking, metadata-aware retrieval, and source attribution for high-accuracy grounded responses.
- Developed scalable PDF/arXiv ingestion system with PyMuPDF, sentence-transformers embeddings, SQLite metadata indexing, and optimized low-latency deployment using lazy loading and Streamlit caching.
- Tags:** GenAI • Advanced RAG • Hybrid Retrieval • LLM Engineering • Retrieval Optimization • Qdrant • Groq • Vector Databases • Semantic Search • Citation Grounding • Research AI • Production AI Systems • Multi-Document Reasoning • AI Infrastructure • Applied AI Engineering

AI Career Copilot — Production-Grade GenAI Career Intelligence Platform

2026

- Built a full-stack AI Career Copilot using Retrieval-Augmented Generation (RAG), semantic search, FAISS vector retrieval, and Groq Llama 3 models to deliver resume-aware job matching, recruiter fit scoring, skill-gap analysis, mock interviews, and personalized learning roadmaps.
- Engineered an end-to-end RAG pipeline with PDF parsing, semantic chunking, sentence-transformer embeddings, FAISS vector indexing, and context-aware retrieval for intelligent resume Q&A and job analysis.
- Developed scalable full-stack architecture using FastAPI, Streamlit, SQLite, JWT authentication, and modular AI workflows optimized for low-latency inference and production-style deployment.
- Tags:** GenAI • Advanced RAG • LLM Engineering • Semantic Search • Vector Databases • FAISS • Groq • Retrieval Systems • AI Infrastructure • Prompt Engineering • Full-Stack AI • Production AI Systems • Resume Intelligence • Applied AI Engineering • FastAPI • Streamlit • NLP • AI Career Systems

CERTIFICATIONS

- Machine Learning Master Course

Link: [Coding Blocks](#) (03/2025-07/2025)

STRENGTHS

- Clear and Concise Communicator
- Disciplined

ACHIEVEMENTS

- Secured 87.84 percentile in JEE MAIN 2021, placing in top 13% candidates nationwide.
- 4 star rating in Problem Solving and SQL on HackerRank.